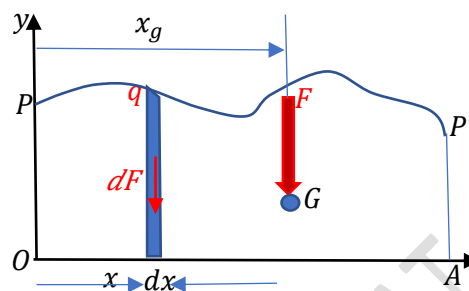


DISTRIBUTED FORCES IN A PLANE

In many situations, we may come across forces distributed over a length, area, or volume. When a force is distributed, it will be specified by its intensity. The intensity of the distributed force is the force per unit length (if it is distributed over a length), force per unit area (if it is distributed over an area), or force per unit volume (if it is distributed over a volume). When a force is distributed over a length, it can be represented by a load-distribution diagram.



Consider any case of a distributed force, acting vertically downwards, represented by the load distribution diagram $OPP'A$. The load diagram is divided into infinitesimal elements, which constitutes a system of elemental parallel forces. Consider one such element over a length dx at a distance x from the origin. Let the intensity of the load be q at this location.

The magnitude of the elemental force, $dF = q \cdot dx = dA$ (the area of the elemental strip)

Total force, $F = \int_0^A dq = \int_0^A dA = \text{Total area of the load distribution diagram}$

The line of action of the resultant force can be found out by Varignon's theorem.

Let the resultant passes through a point G at a distance x_g from O .

Moment of the resultant force about O is equal to the algebraic sum of the moments of the components about the same point.

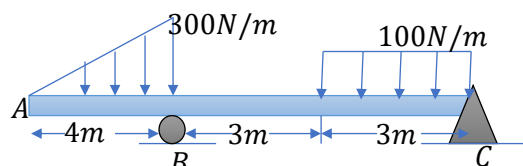
$$F \cdot x_g = \int x \, dF$$

$$x_g = \frac{\int x \, dF}{F} = \frac{\int x \, dA}{\int dA} = \bar{x}$$

That is, the resultant of a distributed force is equal to the area of the force-distribution diagram and its line of action will pass through the centroid of the load-distribution diagram.

Example 1

Determine the support reactions at B and C of the beam loaded and supported as shown.

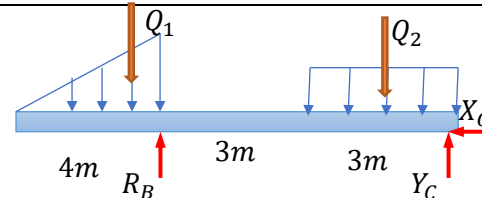


The net force of the uniformly varying load

$$Q_1 = \text{area of the triangle} = \frac{1}{2} \times 4 \times 300 = 600 \text{ N}$$

The net force of the uniformly distributed load

$$Q_2 = \text{area of the rectangle} = 3 \times 100 = 300 \text{ N}$$



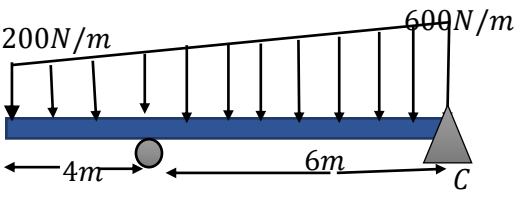
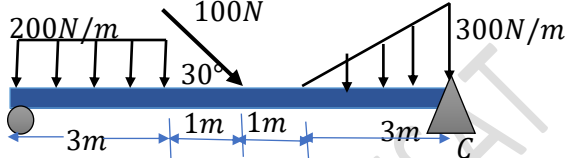
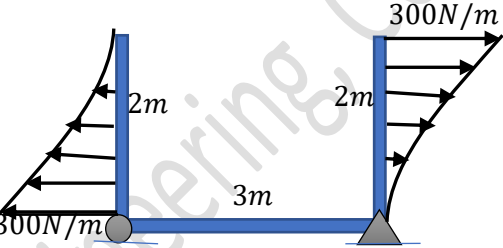
$$\Sigma F_x = 0 \gg \gg X_C = 0$$

$$\Sigma F_y = 0 \gg \gg Y_C + R_B - Q_1 - Q_2 = 0 \gg \gg Y_C + R_B = 900 \text{ N}$$

$$\Sigma M_B = 0 \gg \gg Q_1 \times \frac{4}{3} - Q_2 \times 4.5 + Y_C \times 6 = 0 \gg \gg Y_C = 91.7 \text{ N}$$

$$R_B = 808.3 \text{ N}$$

Exercise 13

13.1 Determine the support reactions of the beam loaded as shown.	
13.2 Determine the support reactions of the beam loaded as shown.	
13.3 Determine the support reactions of the beam loaded with parabolically distributed forces as shown.	

Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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